Orange Problem Assignment

Instructions

1. The questions to be answered will be based on the last three digits of your SRN.
   - Take the last three digits of your SRN.
   - Compute: REM = (Last three digits of SRN) mod 10
   - The remainder (REM) indicates the question number you must attempt.
2. The Orange Problem solutions to be submitted to drive link shared by the course faculty.
3. Submission By: 20th March 2026; Maximum Marks: 10
4. Every question has two sub-questions, both needs to be answered

Question Bank

1. A. Scenario: Military Ration Distribution (6 Marks)

An army unit records the number of ration packets distributed over 7 days.
The data is stored in memory as an integer array.

Given Data

Rations = {120, 150, 100, 180, 130, 170, 140}
N = 7

Problem Statement

Write a RISC-V assembly program using RARS to:

Calculate the total ration packets distributed.
Find the average ration per day (integer division).
Count the number of days where ration distribution is greater than the average.
Determine the maximum ration value.

Constraints

Use loops to access the array.
Use arithmetic and conditional branching instructions.
Do not use pseudo instructions beyond standard RISC-V.
The program must run correctly in RARS.

B. Scenario: Classroom Attendance Analyzer (4 Marks)

An institution records the roll numbers of students present in a class using an integer array.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Count total students present
- Calculate sum and average of roll numbers

- Determine the highest roll number

Constraints

• Use arithmetic instructions only
• Avoid pseudo instructions beyond standard RISC-V
• Program must execute correctly in RARS

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2. A. Scenario: Airport Runway Usage Analysis (6 Marks)

An airport authority records the number of flights that used the main runway over **7** consecutive hours.
The data is stored in memory as an integer array.

Given Data

Flights = {45, 60, 38, 72, 55, 68, 50}
N = 7

Problem Statement

Write a RISC-V assembly program using RARS to:

Calculate the total number of flights recorded.
Compute the average flights per hour (integer division).
Count the number of hours where flight count is below the average.
Determine the minimum number of flights recorded in any hour.

Constraints

Use loops to traverse the array.
Use arithmetic and conditional branch instructions.
Do not use pseudo instructions beyond standard RISC-V.
The program must execute correctly in RARS.

B. Scenario: Mini Bank Account System (4 Marks)

A bank stores the balances of five customers in an integer array.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Calculate total balance
- Find the maximum balance
- Apply 5% bonus to each account and display updated balances

Constraints

• Use arithmetic instructions only
• Avoid pseudo instructions beyond standard RISC-V
• Program must execute correctly in RARS

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3. A. Scenario: Battlefield Sensor Stability Check (6 Marks)

A battlefield sensor records **signal deviation values** every hour.
Positive values indicate noise spikes, negative values indicate signal drops.
The readings are stored in memory as an integer array.

Given Data

Signal = { 5, -3, 12, -7, 9, 0, -2, 15 }
N = 8

Problem Statement

Write a RISC-V assembly program using RARS to:

1. Count the number of stable readings
   (values between -5 and +5, inclusive).
2. Calculate the net signal deviation
   (sum of all positive values minus sum of all negative values).
3. Determine the number of signal drop events
   (values less than 0).
4. Find the first occurrence index of a signal spike
   (value greater than 10).
   If no such value exists, store -1.

Constraints

Use loops to traverse the array.
Use arithmetic, comparison, and conditional branching instructions.
Handle negative numbers correctly.
Do not use pseudo instructions beyond standard RISC-V.
The program must run correctly in RARS.

B. Scenario: Number Property Checker (4 Marks)

Given a positive integer N stored in memory.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Check whether N is even or odd
- Check whether N is prime
- Check whether N equals factorial of any integer

Constraints

• Use arithmetic instructions only
• Avoid pseudo instructions beyond standard RISC-V
• Program must execute correctly in RARS

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4. A. Scenario: Food Plate Chaos (6 Marks)

A restaurant has N different toppings for a salad plate.
The chef is *extremely particular* and prepares a plate for every possible order in which the toppings can be added.

Each topping is used exactly once in a plate.
The number of toppings N is stored in memory.

Given Data

N = 5

Problem Statement

Write a RISC-V assembly program using RARS to:

Determine the total number of unique salad plates the chef can create.
Find the factorial of the total number of unique biryani plates.

Constraints

Use loops to perform calculations.
Use only arithmetic and conditional branch instructions.
Do not use precomputed values or math libraries.
Do not use pseudo instructions beyond standard RISC-V.
The program must run correctly in RARS.

B. Scenario: Sensor Temperature Analysis (4 Marks)

A sensor records temperature readings over 7 intervals stored in an integer array.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Find maximum temperature
- Find minimum temperature
- Compute average temperature
- Count readings above average

Constraints

• Use arithmetic instructions only
• Avoid pseudo instructions beyond standard RISC-V
• Program must execute correctly in RARS

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5. A. Scenario: WhatsApp Sticker Disaster (6 Marks)

A student has N different WhatsApp stickers saved on their phone and wants to spam their friend.
The student considers two different situations:

1. Sticker Order Matters Mode
   The friend gets annoyed if the same stickers arrive in a different order.
2. Sticker Order Doesn't Matter Mode
   The friend only cares about which stickers are sent, not the order.

The value of N and the number of stickers sent R are stored in memory.

Given Data

N = 6
R = 3

Problem Statement

Write a RISC-V assembly program using RARS to:

1. Calculate the total number of different sticker messages possible when order matters.
2. Calculate the total number of different sticker messages possible when order does not matter.
3. Store both results in memory.
4. Display both values using system calls.

Constraints

Use loops for all calculations.
Use only arithmetic and conditional branch instructions.
Do not use any precomputed formulas directly.
Do not use pseudo instructions beyond standard RISC-V.
The program must run correctly in RARS.

B. Scenario: Simple Data Encryption (4 Marks)

A message is stored as an integer array.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Add index value to each element
- Reverse the array
- Display encrypted output

Constraints

• Use arithmetic instructions only
• Avoid pseudo instructions beyond standard RISC-V
• Program must execute correctly in RARS

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6 A. Scenario: Mirror Shop Reality Check (6 Marks)

A mirror shop has a strange rule:
A product code is considered “Mirror-Approved” if it looks exactly the same in the mirror.
If the code changes when seen in the mirror, the shopkeeper rejects it instantly.
The product code is stored in memory as a null-terminated string.

Given Data

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Code = "radar"
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Problem Statement

Write a RISC-V assembly program using RARS to:

Check whether the given product code is Mirror-Approved.
Print “Mirror-Approved” if the code is a palindrome.
Print “Mirror-Rejected” otherwise.

Constraints

Use loops to traverse the string.
Use pointer arithmetic to access characters.
Do not create a reversed copy of the string.

Compare characters from the start and end moving inward.
Use only standard RISC-V instructions.
The program must run correctly in RARS.

B. Scenario: Digital Stopwatch Simulation (4 Marks)

A stopwatch starts from 0 seconds and runs till a given value N.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Display time at each second
- Stop execution when count reaches N

Constraints

• Use arithmetic instructions only
• Avoid pseudo instructions beyond standard RISC-V
• Program must execute correctly in RARS

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7. A. Scenario: Crime Scene Laser Grid (6 Marks)

A high-security vault is protected by a laser grid.
The grid forms a square pattern of size N × N, where lasers are active only on the main diagonal of the grid.
The remaining cells are safe zones.

Given Data

N = 5

Problem Statement

Write a RISC-V assembly program using RARS to print the laser grid pattern such that:

- Print # for laser beams
- Print . for safe zones

Pattern Rule

- A laser (#) is printed when the row index equals the column index
- Otherwise, print a dot (.)

Constraints

Use nested loops (row and column).
Use index comparison logic.
Use only standard RISC-V instructions.
Print characters using RARS system calls.
The program must run correctly in RARS.

B. Scenario: Vending Machine Change Calculator (4 Marks)

Given the price of an item and amount paid.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Calculate change

- Display minimum number of notes/coins

Constraints

• Use arithmetic instructions only

• Avoid pseudo instructions beyond standard RISC-V

• Program must execute correctly in RARS

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8. A. Scenario: Buffet Counter Panic (6 Marks)

A self-service buffet has **limited food counters**, each identified by a **counter number**. Every time a customer visits a counter, the counter number is logged in an array. However:

- Customers may visit the same counter multiple times
- Some counters may get ignored completely

The restaurant head wants to analyze customer behavior.

Given Data

Visits = {2, 4, 2, 1, 5, 2, 4, 3, 4, 4, 1, 2}

N = 12

Problem Statement

Write a RISC-V assembly program using RARS to:

1. Identify the most visited counter number.
2. Identify the least visited counter number among the counters that were visited at least once.
3. Count how many counters were visited more than twice.
4. Print all three results.

Constraints

Use nested loops for frequency counting.
Use only standard RISC-V arithmetic and branch instructions.
Do not use additional arrays or maps.
If there is a tie, select the smallest counter number.
The program must execute correctly in RARS.

B. Scenario: Password Strength Validator (4 Marks)

A numeric password is stored as an integer.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Check for at least one even digit
- Check for at least one odd digit
- Check if sum of digits exceeds threshold

Constraints

• Use arithmetic instructions only

• Avoid pseudo instructions beyond standard RISC-V

• Program must execute correctly in RARS

9. A. Scenario: Alien Pizza Destiny (6 Marks)

An alien machine finds a **destiny number** for every pizza order.
Starting with an integer order number:

- Add all its digits.
  Repeat until a single digit remains.

Given Data

OrderNumber = 987654

Problem Statement

Write a RISC-V assembly program using RARS to:

1. Compute the final single-digit destiny number using repeated digit reduction.
2. Using the destiny number, calculate the factorial of the digit.

Constraints

Use loops and integer arithmetic only. Do not convert the number to a string.
No mathematical shortcuts. The program must run correctly in RARS.

B. Scenario: Lift Movement Simulator (4 Marks)

A lift starts at floor 0 and receives a sequence of movement requests.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Calculate final floor position
- Calculate total floors traveled

Constraints

• Use arithmetic instructions only
• Avoid pseudo instructions beyond standard RISC-V
• Program must execute correctly in RARS

0. A. Scenario: Hospital Medicine Scheduling (6 Marks)

A hospital has two medicines that need to be administered at regular intervals:

- Medicine A every X hours
- Medicine B every Y hours

The hospital wants to plan:

1. How often both medicines **coincide** (i.e., same time for both).
2. How long until a **full repeat cycle** of the combined schedule.

Given Data

X = 6 # hours for Medicine A
Y = 8 # hours for Medicine B

Problem Statement

Write a RISC-V assembly program using RARS to:

1. Compute the GCD of X and Y using loops.
2. Compute the LCM of X and Y using loops and arithmetic.
3. Store both results in memory.

Constraints

Use loops and arithmetic instructions only. Use conditional branching for comparisons. Do not use any library functions or pseudo-instructions beyond standard RISC-V.

The program must execute correctly in RARS.

B. Scenario: Electricity Bill Generator (4 Marks)

Units consumed are stored as an integer value.

Problem Statement

Write a RISC-V assembly program using RARS to:

- Calculate bill based on slab rates
- Display total bill amount

Constraints

• Use arithmetic instructions only
• Avoid pseudo instructions beyond standard RISC-V
• Program must execute correctly in RARS